# EC2 – 50 Major Scenario-Based Questions

## 1–10: Basic & Troubleshooting Scenarios

1. Your application is running on a single EC2 instance and traffic suddenly increases causing the server to slow down. How would you design a scalable solution?
2. You launched an EC2 instance but cannot SSH into it. What troubleshooting steps would you follow?
3. Your EC2 instance is running but the website is not accessible through the browser. What could be the reasons?
4. An EC2 instance status check failed in AWS console. How would you troubleshoot it?
5. Your EC2 instance storage is full and the application has stopped working. How would you fix it?
6. Your EC2 instance suddenly stopped responding to requests. How would you identify the root cause?
7. Your EC2 instance was accidentally terminated. What preventive measures could you implement to avoid this in the future?
8. You launched an EC2 instance but it cannot access the internet. What could be the reasons?
9. Your EC2 instance CPU utilization is consistently above 90%. What actions would you take?
10. Your application deployed on EC2 is consuming too much memory and crashes frequently. How would you resolve this?

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# 11–20: Networking Scenarios

11. Two EC2 instances in the same VPC cannot communicate with each other. What could be the possible reasons?
12. Your EC2 instance is in a private subnet but needs to download updates from the internet. How would you enable this?
13. Users cannot access the EC2 application through port 8080. What would you check?
14. Your EC2 instance has a public IP but the application is still not reachable from the internet. What could be wrong?
15. Your EC2 instance cannot connect to an RDS database. What checks would you perform?
16. Your EC2 instance cannot resolve DNS names. What could be the cause?
17. You want to allow SSH access only from your office network. How would you implement it?
18. Your EC2 instance needs to securely access another EC2 instance in a different VPC. How would you design this?
19. You want to expose an application running on EC2 to the internet securely. What AWS service would you use?
20. Your EC2 instances need to communicate with on-premises servers. How would you achieve this?

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# 21–30: Performance & Scaling Scenarios

21. Your EC2 instance becomes slow during peak traffic hours. How would you design a scalable architecture?
22. Your application requires automatic scaling when CPU utilization reaches 70%. How would you implement it?
23. You want to distribute incoming traffic across multiple EC2 instances. What service would you use?

24. Your EC2 instance disk I/O is very high and affecting application performance. How would you fix it?

25. Your application requires high availability across multiple Availability Zones. How would you design the architecture?

26. You want unhealthy EC2 instances to be automatically replaced. How would you configure this?

27. Your EC2 instance type does not provide enough performance. What options do you have?

28. You want to reduce downtime during application deployment to EC2 instances. How would you achieve this?

29. Your EC2 instances need to scale based on request count instead of CPU usage. How would you configure this?

30. Your application must run with minimal latency for global users. How would you design the infrastructure?

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# 31–40: Security Scenarios

31. Your EC2 instance was accessed by an unauthorized user. What immediate actions would you take?

32. Developers need temporary access to EC2 instances but you don't want to share SSH keys. What solution would you use?

33. You want EC2 instances to access S3 without storing access keys in the server. How would you implement it?

34. You want to restrict EC2 instance access to specific ports only. How would you configure it?

35. Your organization requires all EC2 instances to have encrypted storage. How would you enforce this?

36. You want to monitor suspicious login attempts on EC2 instances. What tools would you use?
37. A developer accidentally deleted important data on an EC2 instance. How would you recover it?
38. You want to block a specific IP address from accessing your EC2 application. How would you implement it?
39. You want to prevent users from terminating EC2 instances. What AWS feature would you use?
40. Your company requires centralized logging for all EC2 instances. How would you implement it?

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# 41–50: Advanced Real-World Scenarios

41. Your company wants to reduce EC2 costs for long-running workloads. What pricing options would you recommend?
42. Your application traffic varies significantly throughout the day. How would you optimize EC2 costs?
43. You need to deploy identical EC2 instances across multiple environments (dev, test, prod). How would you automate it?
44. Your EC2 instances need to be automatically configured after launch. What AWS service or method would you use?
45. Your EC2 instance unexpectedly terminated. How would you investigate the root cause?
46. You want to quickly create multiple EC2 instances with the same configuration. What AWS feature would you use?
47. Your application must run even if one Availability Zone fails. How would you design the architecture?
48. Your EC2 instances need to store large files that exceed local disk capacity. What AWS service would you integrate?

49. You want to automatically back up EC2 instances regularly. What solution would you implement?

50. Your EC2 infrastructure needs to be reproducible and version controlled. What DevOps tools would you use?

# AWS S3 – 50 Scenario-Based Questions

## 1–10: Basic & Usage Scenarios

1. Your application needs to store user-uploaded images and videos. How would you design a solution using S3?

2. Your website has many static files (HTML, CSS, JS). How would you host them using S3?

3. Your EC2 instance disk is filling up due to logs. How would you use S3 to solve this problem?

4. You want users to upload files directly to S3 without passing through your application server. How would you implement this?

5. Your application stores large files in S3 and users want faster download speeds. What solutions would you use?

6. Your development team accidentally deleted files from an S3 bucket. How would you prevent data loss in the future?

7. Your company needs to store backup files for many years. How would you design a cost-effective solution using S3?

8. Your application requires storing millions of files. How would you manage and organize them in S3?

9. Your team wants to migrate files from an on-premise server to S3. What approaches can you use?

10. Your application needs temporary access to upload files to S3. How would you provide secure access?

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# 11–20: Security Scenarios

11. You want only specific IAM users to access an S3 bucket. How would you implement this?
12. Your company policy requires all S3 data to be encrypted. How would you enforce encryption?
13. Your S3 bucket was accidentally made public. How would you prevent this from happening again?
14. You want to allow public access only to specific objects in an S3 bucket. How would you configure this?
15. Your EC2 instances need to access S3 without storing access keys. What solution would you use?
16. Your organization wants to monitor who accessed objects in S3. What AWS services would you use?
17. You want to restrict S3 bucket access only from a specific VPC. How would you implement this?
18. Your company wants to prevent deletion of critical files in S3. What feature would you enable?
19. You want to block all public access to S3 buckets in your AWS account. How would you configure this?
20. Your security team wants to detect unusual activity in S3 buckets. What monitoring tools would you use?

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# 21–30: Performance & Data Management Scenarios

21. Your application downloads files from S3 but the performance is slow for global users. What solution would you use?

22. Your company stores large files in S3 and wants faster upload speeds. What techniques would you use?
23. Your S3 bucket contains millions of objects and searching for files is difficult. How would you organize them?
24. You want to automatically move older files to cheaper storage. What feature would you use?
25. Your company wants to replicate S3 data to another AWS region. How would you configure this?
26. Your application requires high availability for stored data. How does S3 ensure this?
27. Your team needs to process files automatically when they are uploaded to S3. How would you design this?
28. Your application stores frequently accessed files in S3 and wants faster access. What storage class would you use?
29. Your company stores archival data in S3 that is rarely accessed. What storage class would you choose?
30. You want to automatically delete old log files from an S3 bucket. How would you configure this?

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# 31–40: Backup & Disaster Recovery Scenarios

31. Your company needs daily backups of application data to S3. How would you automate this?
32. Your company wants to replicate S3 data to another region for disaster recovery. How would you implement this?
33. Your application data stored in S3 was accidentally overwritten. How can you recover the previous version?

34. Your organization wants to ensure backups cannot be deleted accidentally. What feature would you use?

35. Your company wants to restore archived data stored in Glacier. What steps would you follow?

36. Your company requires long-term storage for compliance (7–10 years). Which S3 storage option would you choose?

37. Your organization wants cross-account backup storage in S3. How would you implement this?

38. Your S3 bucket data must be protected against region failure. How would you design the solution?

39. Your team wants automated backup of EC2 data to S3. What tools or services would you use?

40. Your company wants to track changes made to S3 objects. How would you implement this?

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# 41–50: Cost Optimization & Advanced Scenarios

41. Your company's S3 storage cost is increasing rapidly. How would you optimize costs?

42. You want to automatically move infrequently accessed files to cheaper storage tiers. What feature would you use?

43. Your company stores large log files in S3 but wants to reduce storage cost over time. What strategy would you use?

44. Your organization wants detailed reports about S3 storage usage. What AWS tool would you use?

45. Your company wants to reduce data transfer costs for global users accessing S3 content. What solution would you implement?

46. Your application frequently accesses small files stored in S3. How would you improve performance?

47. Your company wants to automatically archive data older than 1 year. How would you configure this?

48. Your DevOps team wants to manage S3 infrastructure using Infrastructure as Code. What tools would you use?

49. Your company wants to audit S3 bucket configurations across all AWS accounts. What AWS service would help?

50. Your organization wants to receive alerts when large amounts of data are uploaded to S3. How would you implement this?

# AWS VPC – 50 Scenario-Based Questions

## 1–10: Basic VPC Design Scenarios

1. Your company wants to host an application on AWS with a secure network architecture. How would you design a VPC for this application?

2. You need to separate production and development environments within AWS networking. How would you design the VPC structure?

3. Your application servers should not be accessible from the internet, but they must communicate with a public load balancer. How would you design this?

4. Your database must be protected from direct internet access. How would you design the network architecture?

5. Your company requires a **3-tier architecture (web, application, database)** in AWS. How would you design the VPC?

6. You need to deploy EC2 instances in multiple Availability Zones for high availability. How would you design the VPC?

7. Your organization wants to host multiple applications in the same AWS account but isolate their networks. What would you do?

8. Your company needs a dedicated network range for AWS resources. How would you choose and configure the CIDR block?
9. You launched a VPC but forgot to configure internet access. What steps would you take to enable it?
10. Your company wants all production servers to run in private subnets. How would you design this?

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# 11–20: Internet & NAT Gateway Scenarios

11. Your EC2 instances in a private subnet need internet access to download updates. How would you implement this?
12. Your public EC2 instance cannot access the internet even though it has a public IP. What could be the reasons?
13. Your private subnet instances need to access external APIs. How would you enable this securely?
14. You want to allow inbound internet traffic only to load balancers, not directly to EC2 instances. How would you design the architecture?
15. Your NAT Gateway suddenly stopped working and private instances lost internet access. How would you troubleshoot?
16. You want to minimize NAT Gateway cost for multiple private subnets. What architecture would you design?
17. Your company requires all internet traffic from private instances to go through a firewall. How would you design this?
18. Your EC2 instance in a public subnet cannot access the internet. What components would you check?
19. Your team wants to allow outbound internet traffic but block inbound traffic for EC2 instances. How would you configure this?
20. Your NAT Gateway is causing high costs. What alternative solution could you consider?

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# 21–30: Security & Access Control Scenarios

21. You want to restrict SSH access to EC2 instances only from your office IP address. How would you implement this?

22. Your EC2 instances should only communicate with specific servers inside the VPC. How would you enforce this?

23. Your company wants to block all traffic except HTTP and HTTPS in the VPC. How would you configure it?

24. Your security team wants to monitor and log all network traffic in the VPC. What AWS service would you use?

25. Your organization wants to detect suspicious network activity in the VPC. What tools would you use?

26. Your company requires centralized security control across multiple VPCs. How would you design this?

27. You want to prevent EC2 instances from accessing certain external IP addresses. How would you implement this?

28. Your VPC network traffic needs to be logged for compliance purposes. How would you enable this?

29. Your company wants to inspect network traffic between subnets. What solution would you use?

30. Your team accidentally exposed a database to the internet. What changes would you implement immediately?

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# 31–40: Connectivity Scenarios

31. Your AWS VPC needs to connect to your on-premises data center. What connectivity options would you use?
32. Your company requires secure communication between two VPCs in different regions. How would you implement it?
33. Two VPCs need to communicate privately within the same region. What AWS feature would you use?
34. Your company has multiple AWS accounts and needs secure network communication between them. How would you design this?
35. Your EC2 instances must access AWS services like S3 without using the internet. What solution would you use?
36. Your VPC needs to communicate with another VPC that has overlapping CIDR ranges. How would you solve this?
37. Your application requires low-latency connectivity between AWS and your data center. What connectivity solution would you use?
38. Your team wants private access to AWS services from the VPC. How would you configure it?
39. Your VPC needs to connect to multiple other VPCs efficiently. What architecture would you use?
40. Your application servers in one VPC need to access databases in another VPC securely. How would you implement it?

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# 41–50: Troubleshooting & Advanced Scenarios

41. Your EC2 instance cannot communicate with another instance in the same subnet. What could be the reasons?
42. Your EC2 instance cannot reach an external website. What VPC components would you check?

43. Your EC2 instance cannot connect to an RDS database inside the same VPC. How would you troubleshoot?
44. Your VPC suddenly lost internet connectivity. What components would you investigate?
45. Your route table configuration is incorrect and traffic is not reaching the internet. How would you fix it?
46. Your EC2 instance cannot access an S3 bucket even though it has internet access. What could be wrong?
47. Your application latency increased due to network issues. How would you diagnose the problem?
48. Your company wants automated VPC creation across multiple environments. What tools would you use?
49. Your DevOps team wants to standardize VPC architecture for all environments. How would you implement this?
50. Your organization wants to manage network infrastructure using Infrastructure as Code. What tools would you use?

# AWS RDS – 50 Scenario-Based Questions

## 1–10: Basic RDS Usage Scenarios

1. Your application requires a managed relational database without handling OS maintenance. How would you use AWS RDS for this?
2. Your application database is running on EC2 but requires frequent maintenance and backups. How would migrating to RDS help?
3. Your company wants a highly available database for production. How would you configure RDS to achieve this?
4. Your developers want a separate database environment for testing without affecting production. How would you design this?
5. Your application needs automatic database backups. How would you configure it in RDS?

6. Your company wants a database that automatically handles patching and maintenance. How does RDS help?
7. Your application needs to store structured relational data. Which RDS database engine would you choose and why?
8. Your development team needs a temporary database for testing. How would you create and manage it in RDS?
9. Your company wants minimal downtime during database maintenance. How would RDS handle this?
10. Your application requires automatic scaling of storage capacity. How would you configure this in RDS?

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# 11–20: High Availability & Disaster Recovery Scenarios

11. Your primary database server fails unexpectedly. How would RDS ensure high availability?
12. Your company requires zero or minimal downtime during database failure. What RDS feature would you enable?
13. Your application must remain available even if an Availability Zone fails. How would you configure RDS?
14. Your company wants a disaster recovery strategy for database failures across regions. How would you implement it?
15. Your database performance is critical and must support failover. What RDS configuration would you choose?
16. Your organization requires database replication for read-heavy workloads. What RDS feature would you use?
17. Your company wants to replicate data to another AWS region for disaster recovery. How would you achieve this?

18. Your application database crashed and you need to restore it to a specific point in time. How would you do it?

19. Your team accidentally deleted important data from the database. How can you recover it in RDS?

20. Your organization wants automated database snapshots for backup purposes. How would you configure them?

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# 21–30: Performance & Scaling Scenarios

21. Your RDS database CPU utilization is very high and causing slow queries. How would you troubleshoot this?

22. Your application has heavy read traffic causing database performance issues. What solution would you implement?

23. Your database queries are slow and affecting application performance. What steps would you take to investigate?

24. Your database storage is almost full. How would you increase storage capacity in RDS?

25. Your application traffic increased significantly and the database cannot handle the load. What options do you have?

26. Your company needs to improve database read performance globally. What architecture would you design?

27. Your database is handling many concurrent connections and causing performance issues. How would you optimize it?

28. Your application requires low-latency database access across regions. How would you design the solution?

29. Your database workload increased unexpectedly. How would you scale the RDS instance?

30. Your team wants database performance monitoring. What AWS tools would you use?

## 31–40: Security Scenarios

31. Your company wants to ensure only specific EC2 instances can access the RDS database. How would you configure this?

32. Your organization requires all database data to be encrypted. How would you implement encryption in RDS?

33. Your company wants to restrict database access to private networks only. How would you configure RDS?

34. Your team accidentally exposed the database to the internet. What steps would you take to secure it?

35. Your security team wants to audit database access. What monitoring tools would you use?

36. Your company requires secure credentials management for database access. What AWS service would help?

37. Your organization wants database connections to use SSL encryption. How would you enforce this?

38. Your company requires role-based access control for database users. How would you implement it?

39. Your database contains sensitive data and must comply with security regulations. What security features would you enable?

40. Your organization wants to monitor suspicious database activities. What AWS services could help?

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## 41–50: Backup, Cost Optimization & Advanced Scenarios

41. Your company wants daily backups of the database without manual intervention. How would you configure this?

42. Your database storage costs are increasing rapidly. How would you optimize costs?

43. Your team wants to archive old database data to reduce storage costs. What strategies would you use?

44. Your organization wants to automate database provisioning across environments. What tools would you use?

45. Your RDS instance suddenly stopped working. How would you investigate the root cause?

46. Your team wants to deploy the same database infrastructure across dev, test, and production environments. How would you automate it?

47. Your company wants centralized monitoring for database metrics. What AWS service would you use?

48. Your application requires automatic scaling of database capacity based on demand. What RDS options would you consider?

49. Your organization wants automated patch management for databases. How does RDS handle this?

50. Your DevOps team wants to manage database infrastructure using Infrastructure as Code. What tools would you use?

# AWS IAM – 50 Scenario-Based Questions

## 1–10: Basic IAM Scenarios

1. Your company has multiple developers who need access to AWS resources. How would you manage their access using IAM?

2. A new employee joins your team and needs limited access to AWS services. How would you grant the appropriate permissions?

3. Your organization wants to group users with similar permissions together. What IAM feature would you use?

4. Your company wants to give read-only access to some users for monitoring resources. How would you implement it?

5. A developer accidentally deleted an important EC2 instance due to excessive permissions. How would you prevent this in the future?

6. Your company wants to follow the **principle of least privilege**. How would you implement this using IAM?

7. Your organization wants to disable root account usage for daily operations. How would you enforce this?

8. Your team wants to allow developers to view EC2 instances but not terminate them. How would you configure this?

9. Your company requires different access levels for developers, testers, and administrators. How would you design IAM policies?

10. Your organization wants centralized user management for AWS access. What IAM approach would you use?

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# 11–20: IAM Roles Scenarios

11. Your EC2 instance needs access to an S3 bucket without storing AWS credentials. How would you implement this?

12. Your application running on EC2 needs to access DynamoDB securely. What IAM solution would you use?

13. Your company wants to allow Lambda functions to access S3 buckets. How would you configure this?

14. Your organization has applications running on EC2 that need temporary AWS access. What IAM feature would you use?

15. Your team wants to allow users from another AWS account to access resources in your account. How would you implement it?

16. Your company wants cross-account access between two AWS accounts. What IAM configuration would you use?

17. Your application requires temporary permissions to access AWS resources. What service would provide this?

18. Your EC2 instances need access to multiple AWS services securely. What IAM solution would you implement?

19. Your organization wants to allow Kubernetes pods to access AWS services securely. What IAM feature would you use?

20. Your DevOps team wants automated services to interact with AWS resources securely. What IAM mechanism would help?

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# 21–30: Security & Access Control Scenarios

21. Your company wants to enforce **Multi-Factor Authentication (MFA)** for all IAM users. How would you implement it?

22. Your organization wants to detect unusual login attempts to AWS accounts. What AWS services would you use?

23. Your company wants to restrict access to AWS resources based on IP address. How would you configure this?

24. Your team accidentally exposed AWS access keys in a GitHub repository. What immediate actions would you take?

25. Your organization wants to rotate IAM access keys regularly. How would you manage this?

26. Your company wants to block access to AWS resources outside office hours. How would you implement this?

27. Your security team wants to prevent public access to sensitive AWS resources. How would you enforce this using IAM?
28. Your company wants to audit who accessed AWS resources. What monitoring services would you use?
29. Your organization wants to detect overly permissive IAM policies. What tools would help?
30. Your company requires strict access control for production resources. How would you enforce this?

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# 31–40: Monitoring & Compliance Scenarios

31. Your company wants to track every API call made in the AWS account. What service would you enable?
32. Your organization needs a report showing all IAM users and their permissions. How would you generate it?
33. Your company wants to detect unused IAM roles and policies. What AWS tool would help?
34. Your organization wants to monitor failed login attempts to AWS accounts. What services would you use?
35. Your security team wants alerts when IAM policies are modified. How would you configure this?
36. Your organization needs compliance checks for IAM policies. What AWS service would help?
37. Your company wants to ensure IAM users do not have long-term credentials. How would you enforce this?
38. Your team wants to review IAM access permissions regularly. What AWS tools would help?

39. Your organization wants automated alerts when new IAM users are created. How would you implement this?

40. Your company requires periodic audits of IAM roles and policies. What services would you use?

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# 41–50: Advanced & Real-World DevOps Scenarios

41. Your company wants to manage IAM infrastructure using Infrastructure as Code. What tools would you use?

42. Your DevOps team wants automated IAM role creation during infrastructure deployment. How would you implement this?

43. Your company wants to standardize IAM roles across multiple AWS accounts. How would you design this?

44. Your organization wants to allow federated login from corporate Active Directory. How would you implement this?

45. Your company wants employees to log in to AWS using company credentials. What solution would you use?

46. Your organization wants centralized IAM management across multiple AWS accounts. What AWS service would help?

47. Your company wants to enforce permission boundaries for IAM users. How would you implement this?

48. Your team accidentally created an IAM policy with full administrative access. How would you detect and fix this?

49. Your organization wants automated security checks for IAM configurations. What tools would you use?

50. Your company wants to restrict AWS service access based on resource tags. How would you configure this?